

Research Tools: LaTeX Libraries Demonstration

Asterinos

September 24, 2026

1 Introduction / Εισαγωγή

This document demonstrates the two custom packages in the `latex/` folder:

- `greekenglish.sty`: Enables typing both English and Greek seamlessly using Unicode fonts with LuaLaTeX / XeLaTeX.
- `code.sty`: Provides stylish dark-mode syntax highlighted code environments for various languages.

1.1 Bilingual Text Switching (Άμεση εναλλαγή γλωσσών)

Here is an example use of English and Greek text:

The quick brown fox jumps over the lazy dog. Η γρήγορη καφέ αλεπού πηδά πάνω από το τεμπέλικο σκυλί. Η επιστήμη των υπολογιστών και οι σύγχρονες τεχνολογίες επεξεργασίας δεδομένων. Computer science and modern data engineering technologies.

No commands were required.

2 Code Highlighting Examples

2.1 Python Environment

```
1 # Python sample code
2 def fibonacci(n: int) -> list[int]:
3     """Generate Fibonacci sequence up to n numbers."""
4     if n <= 0:
5         return []
6     seq = [0, 1]
7     while len(seq) < n:
8         seq.append(seq[-1] + seq[-2])
9     return seq[:n]
10
11 if __name__ == "__main__":
12     print(f"Fibonacci(8): {fibonacci(8)}")
```

2.2 SQL Environment

```
1 -- Query user research analytics
2 SELECT
3     u.id,
4     u.username,
5     COUNT(r.id) AS total_runs,
```

```
6      AVG(r.execution_time_ms) AS avg_time_ms
7 FROM users u
8 LEFT JOIN research_runs r ON u.id = r.user_id
9 WHERE r.created_at >= '2026-01-01'
10 GROUP BY u.id, u.username
11 ORDER BY total_runs DESC;
```

2.3 JavaScript Environment

```
1 // Modern async fetch function
2 async function fetchResearchData(endpoint) {
3     try {
4         const response = await fetch(`https://api.research.local/${endpoint}
5         });
6         if (!response.ok) {
7             throw new Error(`HTTP error! status: ${response.status}`);
8         }
9         const data = await response.json();
10        return data;
11    } catch (err) {
12        console.error("Failed to fetch research metrics:", err);
13    }
14 }
```